

SHIVANI KAPASE

Junior Software Developer | Full-Stack Developer (MERN) | AI & Machine Learning

Pune, Maharashtra, India | +91-9370446447 | shivanikapase755@gmail.com | linkedin.com/in/shivani-kapse-54b513309 | github.com/ShivaniKapase643 | shivani-dev-1.netlify.app

PROFESSIONAL SUMMARY

Software Developer and B.E. Computer Engineering student (SGPA 9.6/10) with 4+ months of internship experience building full-stack applications using Java, JavaScript, React.js, Node.js, Express.js, MongoDB, and MySQL, alongside ongoing applied machine learning research at MES Wadia College of Engineering. Passionate about AI-driven software development — independently built several projects combining full-stack engineering with machine learning and generative AI, including an autonomous multi-agent examination platform, a Gemini-powered sustainability assistant, and an AI-based career-prep tool. Strong foundation in Data Structures & Algorithms and Git-based Agile workflows. Seeking a Junior Developer role to grow as a full-stack engineer while contributing to AI-driven product development.

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, SQL
- **Web & Full-Stack:** React.js, Node.js, Express.js, HTML5, CSS3, MERN Stack, RESTful API Design & Integration, JWT Authentication
- **AI/ML Foundations:** Python (NumPy, Pandas, Scikit-learn), Machine Learning Basics, Collaborative Filtering, Working Knowledge of LLM APIs (OpenAI, Gemini)
- **Databases:** MySQL, MongoDB, SQL Query Design, Database Schema Design
- **Data Structures & Algorithms:** Arrays, Linked Lists, Trees, Graphs, Sorting, Dynamic Programming
- **Tools & Practices:** Git, GitHub, VS Code, Postman, Agile, Scrum, Code Reviews, Sprint Planning, Version Control

WORK EXPERIENCE

Research Science Intern (Machine Learning)

2025 – Present

MES Wadia College of Engineering, Pune, India

- Conducting applied machine learning research under faculty mentorship, training and evaluating models in Python using Scikit-learn and Pandas.
- Built data preprocessing, feature engineering, and model evaluation pipelines; studied the statistics and machine learning foundations behind model behavior.
- Documented experiments and maintained reproducible, version-controlled code on GitHub.

MERN Stack Developer Intern — Sunbeam Infotech Pvt. Ltd., Pune, India

Jan 2026 – Feb 2026

- Developed a role-based Online Course Portal using MongoDB, Express.js, React.js, and Node.js, enabling admin course management and student access to recorded lectures for 100+ active users.
- Designed and normalized MySQL and MongoDB database schemas, integrating 8 RESTful APIs that reduced average data retrieval time by 30% and improved frontend-backend communication.
- Built 10 responsive React.js UI components with reusable design patterns, improving page load performance by 25% across mobile and desktop.
- Implemented JWT-based authentication and authorization middleware, securing sensitive student and admin data across the application.
- Collaborated with an Agile team of 5 developers using Git for version control, participating in daily stand-ups, code reviews, and sprint planning, delivering all assigned features on schedule.

Android Development Intern — Microdynamic Software Pvt. Ltd., Pune, India

Jun 2023 – Jul 2023

- Developed and tested 3 Android application modules in Java using Android Studio, delivering all modules within a 6-week timeline with zero critical defects at launch.
- Designed UI components following Google Material Design guidelines, ensuring accessibility compliance across 10+ screen sizes.
- Identified and resolved 15 bugs through systematic debugging and code reviews, reducing the application crash rate by 40%.
- Integrated REST API endpoints for real-time data synchronization between the mobile app and backend server, reducing data latency by 50%.

AI & FULL-STACK PROJECTS

DRONA AI — Autonomous Multi-Agent Examination Integrity Platform | FastAPI, React, TypeScript, PostgreSQL, CrewAI, WebSockets, OpenAI Vision, MediaPipe

Live: drona-ai-rho.vercel.app

- Architected a 6-agent autonomous AI system for online examination integrity covering proctoring, fraud detection, paper generation, analytics, alerting, and fairness auditing.
- Implemented a two-stage AI proctoring pipeline using MediaPipe FaceMesh (468 landmarks) and OpenAI Vision, reducing cloud inference cost via local-first anomaly screening.

GreenChain — AI-Powered Green Logistics & Sustainability Platform | *React, Vite, FastAPI, Gemini AI, Supabase, Pathway*

Live: greenchaina.netlify.app

- Built a platform tracking per-shipment carbon footprint and calculating Gold Standard carbon credits against India's national emission baseline.
- Developed 7 predictive ML models (CO₂ prediction, anomaly detection, driver profiling, route recommendation, green-grade scoring) surfaced through a unified dashboard.

SmartHire AI — AI Placement Preparation Platform | *React, Node.js, MongoDB, Redis, Socket.io, Gemini AI*

Live: hireproai.netlify.app

- Designed an AI-based career-prep platform using a microservices architecture with Redis caching to keep API responses fast under concurrent load.
- Developed AI mock interviews, ATS resume scoring, and job-tracking modules with real-time analytics dashboards using Socket.io and React.

Movie Recommendation System | *Python, Scikit-learn, Pandas, NumPy, BeautifulSoup*

- Engineered an AI-powered recommendation engine using collaborative filtering on a 5,000+ movie dataset, achieving 85% prediction accuracy.
- Built web scraping and preprocessing pipelines to collect and clean movie metadata, improving recommendation relevance by 20%.

Smartshetakri — Agricultural Intelligence Platform | *React.js, Node.js, MongoDB, Python, REST APIs*

- Built a data-driven farming platform delivering crop recommendations via 6 RESTful API endpoints, using weather, soil, and yield data for 200+ farmers.
- Designed a responsive React.js dashboard with data visualizations to help farmers interpret recommendations and make faster decisions.

AI Podcast Generator | *React, Node.js, AI Audio Generation*

Live: aipodcastgenerator.netlify.app

- Built an end-to-end platform that generates podcast episodes from a user-selected topic, with accessibility-focused design for audio-first content discovery.
- Implemented playlist creation, listening history, episode sharing, and a scheduling feature for automated recurring episode delivery.

EDUCATION

Bachelor of Engineering in Computer Engineering — MES Wadia College of Engineering, Pune (SPPU)

SGPA: 9.6 / 10.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Diploma in Computer Engineering — Government Residence Women's Polytechnic, Tasgaon (MSBTE)

Percentage: 90.41%

Secondary School Certificate (SSC) — Siddheshwar High School, Pimprad (Maharashtra State Board)

Percentage: 92.60%

CERTIFICATIONS

- MERN Stack Development — Sunbeam Infotech Pvt. Ltd., Pune (Jan 2026 – Feb 2026)
- Java Programming Certification — Core Java and Object-Oriented Programming
- JavaScript Certification — Frontend and Backend JavaScript Fundamentals
- Data Structures Using C — Foundational DSA Concepts and Algorithm Implementation

ACHIEVEMENTS

- Top 15 Finalist — Pune Agri Hackathon International 2026, competing against 500+ teams from multiple countries.
- Participant — Smart India Hackathon 2025 (Govt. of India) and ISRO Bharatiya Antriksha Hackathon.
- Finalist — Mock Placement Drive, ELECTROVERT 2023-24, Walchand College of Engineering, Sangli.